

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

- 27. How would you present your simulation results to a client? (BL4)(1)
- 28. Explain jitter using a real-world analogy (BL5)(1)
- 29. Describe how you tested network performance in a lab (BL4)(1)
- 30. Explain the importance of performance metrics in one structured answer (BL5)(1)
- 31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
- 32. Why are you interested in networking and simulation tools? (BL5)(1)
- 33. What makes you confident in analyzing network performance? (BL5)(1)
- 34. Explain a concept you are very strong in (related to networking) (BL5)(1)
- 35. How would you handle a question you don't know? (BL5)(1)
- 36. Describe a situation where you solved a technical problem (BL4)(1)
- 37. Why should we select you for a networking role? (BL5)(1)
- 38. What challenges did you face while learning Packet Tracer? (BL5)(1)
- 39. How do you improve your technical skills regularly? (BL5)(1)
- 40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. How does network congestion affect packet delay and packet loss?

Network congestion occurs when too much data enters the network. It increases **packet waiting time**, causing higher delay. When router buffers become full, packets are dropped, resulting in **packet loss**.

2. Compare star and mesh topologies.

Star topology is easier to manage, but failure of the central device can affect the network. **Mesh topology** provides higher reliability and fault tolerance because multiple paths are available, but it is more expensive and complex.

3. Why do more routers increase end-to-end delay?

Every router introduces **processing and queuing delay**. Packets also take time to be transmitted and forwarded from one router to another. Therefore, more routers generally increase total end-to-end delay.

4. How does QoS improve voice and video traffic?

QoS gives priority to delay-sensitive traffic such as **voice and video**. It reduces delay, jitter, and packet loss, resulting in smoother calls and video streaming.

5. Impact of fiber optic instead of copper?

Fiber provides **higher bandwidth, lower signal loss, and longer transmission distances** than copper. It is also less affected by electromagnetic interference, making it suitable for high-speed communication.

6. How do VLANs improve performance and security?

VLANs divide a network into separate logical networks. This **reduces unnecessary broadcast traffic** and improves performance. It also isolates users and helps prevent unauthorized access.

7. Compare TCP and UDP.

TCP is reliable because it provides acknowledgements, sequencing, and retransmission, but it has more overhead. **UDP** is faster and has lower overhead but does not guarantee delivery, making it suitable for applications such as streaming and gaming.

8. Why is load balancing important?

Load balancing distributes network traffic across multiple links or servers. It prevents one resource from becoming overloaded, improving **performance, availability, and reliability**.

9. How do frequent routing updates affect network efficiency?

Frequent routing updates consume **bandwidth and router processing resources**. Excessive updates can reduce network efficiency, although necessary updates help routers maintain accurate routes.

10. Security vs network performance

Firewalls and encryption improve network security but require additional processing. This can slightly increase **delay and resource usage**. Properly configured security mechanisms provide protection while minimizing performance impact.

11. A network has high delay. How will you identify and fix it?

Use tools such as **ping and traceroute** to identify where delay occurs. Check router load, congestion, bandwidth, and link problems. Then optimize routing, reduce congestion, or upgrade the link if necessary.

12. Why is throughput low despite high bandwidth?

Possible reasons include **network congestion, packet loss, high latency, inefficient protocols, hardware limitations, or interference**. High bandwidth alone does not guarantee high actual throughput.

13. How will you reduce packet loss?

Reduce congestion using **traffic management and QoS**, increase available bandwidth, replace faulty cables/devices, and optimize routing. Monitoring helps identify the exact source of packet loss.

14. VoIP call is breaking frequently. How will you troubleshoot?

Check **packet loss, delay, jitter, and bandwidth** using network monitoring tools. Verify the router, connection, and QoS configuration. Give VoIP traffic higher priority if necessary.

15. How can high jitter be improved?

Use **QoS, traffic prioritization, sufficient bandwidth, and better routing**. Reducing congestion and packet queuing helps maintain a consistent packet arrival time.

16. How will you simulate congestion in Packet Tracer?

Create multiple devices generating traffic through a common link. Reduce the link bandwidth or increase traffic load and observe **packet delay, packet loss, and throughput** using simulation mode.

17. Packets are delayed in a router. What steps will you take?

Check the router's **CPU utilization, queue size, bandwidth, and configuration**. Use ping/traceroute to locate the problem. Reduce traffic, optimize routing, or upgrade the router/link if required.

18. How would you optimize a network for online gaming?

Provide **low latency, low jitter, and low packet loss**. Use QoS to prioritize gaming traffic, avoid congestion, choose efficient routes, and provide sufficient bandwidth.

19. Design a small network and measure its performance.

Create a network with PCs, a switch, and a router. Measure **throughput, delay, packet loss, and jitter** using ping and simulation tools. Compare the measured values with expected performance.

20. If RTT is very high, how will you diagnose it?

Use **ping and traceroute** to identify the source of the delay. Check each link and router for congestion, high utilization, or routing problems. Correct the faulty or overloaded section.

21. Explain network performance to a non-technical person.

Network performance means **how well a network works**. It can be understood by checking how fast data travels, how much data is delivered, how long it takes, and how often data is lost.

22. Describe throughput using a real-life example.

Throughput is the **actual amount of data successfully transferred per second**. For example, if a network successfully transfers 50 MB in one second, its throughput is 50 MB/s.

23. How does packet loss affect video streaming?

Packet loss means some video data does not reach the receiver. This can cause **freezing, buffering, missing frames, or reduced video quality**.

24. Explain delay types.

The main types are **processing delay, queuing delay, transmission delay, and propagation delay**. Together, these determine how long a packet takes to travel through the network.

25. Describe Cisco Packet Tracer professionally.

Cisco Packet Tracer is a **network simulation tool** used to design, configure, test, and troubleshoot network devices virtually without requiring physical hardware.

26. Explain QoS simply and technically.

Simply, QoS means **giving important traffic higher priority**. Technically, QoS manages traffic using prioritization, queuing, and bandwidth allocation to control delay, jitter, and packet loss.

27. How would you present simulation results to a client?

Present the results using **tables, graphs, and key performance metrics** such as throughput, delay, packet loss, and jitter. Explain the problems found and recommend suitable improvements.

28. Explain jitter using a real-world analogy.

Jitter is the **variation in packet arrival time**. For example, if a bus is expected every 10 minutes but arrives after 5, 15, and 8 minutes, the irregular timing is similar to network jitter.

29. How did you test network performance in a lab?

I created and configured the network in **Cisco Packet Tracer**, generated traffic, and measured throughput, delay, packet loss, and other performance metrics. I then analyzed the results.

30. Why are performance metrics important?

Performance metrics help determine whether a network is working efficiently. **Throughput, delay, packet loss, jitter, and bandwidth utilization** help identify problems and guide network improvements.

31. Introduce yourself and explain your network-performance knowledge.

I am a student interested in computer networking. I have learned concepts such as **bandwidth, throughput, delay, packet loss, jitter, QoS, and network simulation using Cisco Packet Tracer**.

32. Why are you interested in networking and simulation tools?

I am interested in networking because it explains how devices communicate and share information. Simulation tools allow me to **design and test networks safely without requiring physical equipment**.

33. What makes you confident in analyzing network performance?

My understanding of **performance metrics and troubleshooting methods** gives me confidence. I can analyze throughput, delay, packet loss, and jitter to identify possible network problems.

34. Explain a networking concept you are strong in.

I am strong in **QoS**. QoS prioritizes important traffic such as voice and video, helping reduce delay, jitter, and packet loss during network congestion.

35. How would you handle a question you don't know?

I would honestly say that I am not completely sure about the answer. I would explain what I know, think logically about the problem, and learn the correct answer afterward.

36. Describe a situation where you solved a technical problem.

While working on a network simulation, if communication failed, I would check **IP addresses, cables, interfaces, routing, and device configuration** step by step until the problem was identified and corrected.